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Measurement of Zonal Flows in a Tokamak Using Langmuir Probe Array

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Abstract

Zonal flows have been measured with several novel Langmuir probe arrays on the HT-7 tokamak since the year 2002. Based on the original idea of the Reynolds stress triple-tip array, we designed the forked probe and its improved generation, the farmer fork probe. And in order to measure the radial shearing rate of zonal flows, we designed the pyramid probe. This paper addresses the technical aspects regarding this new application. Since the zonal flow dynamics is now widely believed crucial to plasma confinement physics, this has opened up an important application possibility for the Langmuir probe.

Langmuir probes are still one of the most popularly applied diagnostics in tokamak plasma edge. This has resulted from the relative simplicity of the method and the spatially localized nature of the measurements. Since the turbulence and anomalous transport are still recognized as one of the most undemonstrated fields in plasma physics, the application of probes to fluctuation measurements often remains to be preferred diagnostics in the plasma edge, although the non-perturbing spectroscopic methods have already become rather powerful. Using an electrical probe to detect fluctuations does not involve the problems in Langmuir probe theory, primarily lying in the interpretation of the electron-side probe voltage-current characteristics in strong magnetic fields. Applying an electrical probe to fluctuation measurements has already become an independent application field for Langmuir probe diagnostics.

The zonal flows have recently been recognized by theorists as a potential regulating factor, possessing possible importance on plasma turbulence and relevant transport. The theoretical effort has been followed by extensive simulation demonstrations. And the predictions from theories and simulations are supported by several recent experiments. (See review paper [1] and references therein) The distinctive features of zonal flows, by which we mean azimuthally symmetric band-like shear flows, are that they can be spontaneously generated by turbulence vie the action of the Reynolds stress and in return suppress turbulence through the E×B decorrelation mechanism. These features give rise to a feed back loop between zonal flows and turbulence, making the plasma turbulence exist in a self-regulated state mediated by zonal flows. So it is expected that the zonal flows/turbulence dynamics control the fluctuation level and, as a result, the turbulent transport. It is well known that zonal flows are a ubiquitous phenomenon in nature, such as the Jupiter belts and zones, and the terrestrial atmospheric jet stream; these are familiar to nearly everyone. The ubiquity of zonal flows in nature makes most plasma physicists believe in the existence of zonal flows in laboratory fusion plasma with enough reasons and confidence. As a result zonal flows physics have recently become the subject of intense interest and investigation in the magnetic confinement plasma field.

(See P.H.Diamond's overview in IAEA2004 [2])

In the context of the tokamak configuration, zonal flows are toroidally and poloidally symmetric, and radially localized potential perturbations. That means only in the radial direction do they have finite wavelength. Therefore, essentially the wave field of zonal flows is just potential fluctuations between nested magnetic flux surfaces. In the strong toroidal magnetic field B_ϕ of the tokamak, the E×B flows will be driven by the time-varying Er and flow mostly in a poloidal direction. That is why we call them zonal flows. The suppressing ability of higher frequency zonal flows is weaker than the low frequency components [3], so the latter are more interesting to us. The generation process of such flows is the radial turbulent transport of poloidal momentum, i.e. the Reynolds stress. This process is the same as the radial turbulent bipolar transport. In this process time-varying radial currents Jr are produced and repeatedly charge the magnetic flux tube, subsequently time-varying Er are built up between nested magnetic flux surfaces and consequently the E×B flows. This generation process is the key of zonal flow physics, and so an experiment to demonstrate the existence of zonal flows must present information on this generation process, in other words, measure the turbulence Reynolds stress. It is well known that the direct measurement of the Reynolds stress in fusion plasmas is not easy. Fortunately, Langmuir probe arrays provide a simple approach. Up to now, besides the Langmuir probe, heavy ion beam probes and several spectroscopic diagnostics have been used for measurements of zonal flows, but only the Langmuir probe can provide direct information on the Reynolds stress. This primarily benefits from the spatially localized nature of the measurement. Although at present the experimental demonstrations of zonal flows are still not sufficient, it has already been widely recognized and accepted that zonal flows are really playing a role in nearly all cases and regimes of laboratory fusion plasma, indeed, so much so that this classic problem has now become a hot topic in the plasma physics research.[1]

To present the direct measurement of the Reynolds stress and zonal flows, we designed a novel probe array, namely, the forked probe array in the year of 2002 (see figure 1). The specially designed probe array has two probe heads poloidally separated by 31.4 mm, which is much longer than the poloidal correlation length $L_{c\theta}$ of turbulence at the plasma edge, which is usually around 1 cm. On each of the two heads a triple tip array is used to provide local measurement of the radial electric field E_r and electrostatic Reynolds stress $\langle \tilde{V}_r \tilde{V}_\theta \rangle$, where $\langle \cdots \rangle$ denotes an ensemble average. All tips are used to measure the floating potential (ϕ_{f1}, ϕ_{f2}, …, ϕ_{f6}). Then the radial and poloidal E_r fluctuations can be calculated as $\tilde{E}_{r,1} = (\tilde{\phi}_{f1} - \tilde{\phi}_{23})/\delta_r$ and $\tilde{E}_{\theta 1} = (\tilde{\phi}_{f2} - \tilde{\phi}_{f3})/\delta_\theta$, ignoring the contribution from electron temperature [4], where $\tilde{\phi}_{23} = (\tilde{\phi}_{f2} + \tilde{\phi}_{f3})/2$. Thus, the poloidal and radial E×B flows are calculated as $\tilde{V}_{\theta 1} = \tilde{E}_{r,1}/B_\phi$ and $\tilde{V}_{r,1} = \tilde{E}_{\theta 1}/B_\phi$. Similarly, the triple tips on the right probe head give $\tilde{V}_{\theta 2} = \tilde{E}_{r,2}/B_\phi$. This probe array structure has the merit that the middle point between tip2

and tip3 is at the same poloidal location as tip1 so that the simply calculated potential $\tilde{\phi}_{23} = (\tilde{\phi}_{f2} + \tilde{\phi}_{f3})/2$ is very close to the floating potential at the middle point between tip2 and tip3, thus this structure will not introduce phase shift within Er measurement. Because calculating the Reynolds stress needs an ensemble average from a large quantity of data, a little phase shift will accumulate a material error and make the measured Reynolds stress meaningless. The probe is mounted on the top of the tokamak along the central line and operated with shot-to-shot scanning. Feedback control of the plasma position ensures that the two heads are located on the same magnetic flux surface. Both the probe tips and the shield tubes are made of graphite to allow the reliable measurement inside the last closed flux surface (LCFS). The data are usually sampled at 1 or 2MHz with 12-bit resolution using a multi-channel digitizer. The boundary profiles of plasma parameters are measured with a fast reciprocating Langmuir probe using a standard triple tip array. All measurements are made in the laboratory setting.

The experiments are carried out on the HT-7 tokamak, which is a superconducting tokamak with two circular poloidal limiters toroidally separated by 180° and a high-field-side belt limiter. In SOL the flows suffer from the obstruction of limiters, so they can only be found in the confinement region. In order to enable the probe to make reliable measurements deeply inside the LCFS, we usually conduct this experiment in low plasma current I_p ohmic heated discharges. The typical parameters are R_0 = 122 cm, a = 27 cm, $B_\phi \cong$ 1.8~2.0 T, $I_p \cong$ 80 kA, $\bar{n}_e \cong 1.5\times10^{19}$ m^{-3}, $T_{e0} \cong$ 0.4 keV, $n_{e\text{-}edge} \cong 1.5\sim4\times10^{18}$ m^{-3}, $T_{e\text{-}edge} <$ 50 eV and discharge duration ≅ 1s. Empirically, such low plasma current can ensure reliable measurement 4 cm inside LCFS. When the probe is inserted 4.5 cm inside LCFS, the secondary electron emission rapidly bursts out after about 100 ms discharge. In such low parameters discharges, the edge safety factor q_a>7 and there is no sawtooth and very little MHD activity. It should be mentioned here that the probe array configuration has a merit in that it can avoid the interference signal from MHD perturbations. Because low mode (m = 2 or 3) MHD perturbations occur in the plasma core and rapidly propagate outwards. The radial distance between tip1 and tip2/tip3 is no more than 5 mm. When MHD perturbations pass through such a short interval, the phase difference and magnitude decay will be very small. The difference between two radial spaced floating potential $\tilde{E}_{r1} = (\tilde{\phi}_{f1} - \tilde{\phi}_{23})/\delta_r$ will offset the MHD perturbations. However, this function does not act on zonal flows, because zonal flows have shear in the radial direction, their small radial scale can be comparable with the radial distance.

High-resolution of poloidal wave number ($-1< k_\theta <1$ rad/cm) can be achieved using this forked probe, so that small k_θ components of poloidal E×B flows can be distinguished from the ambient turbulence. Application of this method to discerning the large-scale flows requires that the lifetime of turbulence eddies is shorter than the transit time of turbulence eddies across the long distance. This condition is satisfied for the specially designed probe. The decorrelation time of floating potential fluctuations at the plasma edge is about $\tau_c \cong 8$ μs. The equilibrium flow velocity is about 2 km/s, resulting in a 16 μs transit time τ_d across the 31.4

mm spatial separation, so that $\tau_c<\tau_d$. The zonal flows are expected to correlate between two points with long poloidal space, while the correlation in ambient turbulence will decay with the increase of distance. Therefore, using the two-point correlation method between the two probe heads, the coherent low frequency flows can be extracted from the background fluctuations.

To measure the radial wave number k_r of this low-frequency mode, one probe head is made 1 cm longer than the other one, so the two probe heads are not only poloidally separated by 31.4 mm but also radially separated by 10 mm (see figure 2). Through the long distance correlation between $\tilde{V}_{\theta 1}$ and $\tilde{V}_{\theta 2}$, the wave number $\bar{k} = \bar{k}_\theta + \bar{k}_r$ with two components is obtained. Since the k_θ of low-frequency zonal flows is about zero, its k_r can be approximately estimated as $k_r \cong k$. The k_r can be used to calculate a turbulence-shearing rate $\omega_s = d\tilde{V}_\theta / dr = 2\tilde{V}_\theta^{pp} / \lambda_r$, where λ_r is the radial wavelength and $\tilde{V}_\theta^{pp}$ is the peak-to-peak value of zonal flows fluctuations.

The zonal flows' generation mechanism has been considered as a three-wave coupling process in which smaller-scale turbulences transfer their energy nonlinearly to larger-scale zonal flows through the Reynolds stress [5], the energy is transferred via the inverse cascading of fluctuation spectra. So the process of zonal flows generation can be investigated using the cross-bicoherence analysis $b^2(f) = \sum_{f=f1+f2} \left\langle \tilde{V}_r(f_1)\tilde{V}_\theta(f_2)\tilde{V}_\theta^*(f) \right\rangle$ [6], namely, that of the radial and poloidal E×B velocities of the turbulence with the zonal flows, where the $\tilde{V}_r$ and $\tilde{V}_\theta$ are directly measured with the forked probe. [7]

The experiment in 2002 only demonstrated the existence of zonal flows at the plasma edge, since theory not only predicted the zonal flows' generation but also predicted that zonal flows could suppress turbulence through the shear decorrelation mechanism, and as a result important to transport, the next step would be researching this shear process. So we improved the forked probe, increased a probe head, as shown in figure 3. The modified probe array was named the farmer fork probe. It has three probe heads, and the middle one toroidally shifts a little to avoid sheltering the poloidal flows. On the top of the middle head a traditional triple tip array was used to measure the radial particle flux $\Gamma_e(t) = \left\langle \tilde{V}_r \tilde{n}_e \right\rangle = \dfrac{\left\langle \tilde{E}_\theta \tilde{n}_e \right\rangle}{B_\phi} \propto \left\langle \tilde{E}_\theta \tilde{I}_s \right\rangle$, where E_θ fluctuations were calculated as $\tilde{E}_\theta = \left(\tilde{\phi}_{f8} - \tilde{\phi}_{f9}\right)/\delta_\theta$, $\tilde{I}_s$ was the ion saturation current fluctuation measured by the center tip7. By researching the correlation between the turbulence-driven particle fluxes and the zonal flows measured by another two probe heads, we could extract the information on the interaction between turbulence transport and zonal flows. The theory predicted that zonal flows could modulate the turbulence flux by the so-called random shear process. The farmer fork probe presented the unique diagnostic function in this process.

The farmer fork probe has been applied to HT-7 for three years, 2003, 2004 and 2005. The

radial shearing rate of zonal flows is very interesting to theorists. However, the direct measurement of shearing rate need at least three points in the radial direction, the farmer fork probe can not do this. So we designed a new probe array in the year 2003, we named it the pyramid probe, as shown in figure 4. This probe array has three layers, with a bilateral symmetry. Using this probe the radial-three-points-floating-potentials measurement can be achieved. The time-dependent radial shearing rate of zonal flows can be simply calculated as $\frac{\partial \tilde{E}_r}{\partial r}(t) = \frac{\tilde{\phi}_1 + \tilde{\phi}_{45} - 2\tilde{\phi}_{23}}{\delta_r^2}$, where $\tilde{\phi}_{45} = (\tilde{\phi}_{f4} + \tilde{\phi}_{f5})/2$ and $\tilde{\phi}_{23} = (\tilde{\phi}_{f2} + \tilde{\phi}_{f3})/2$. Simultaneously tip6 measures the ion saturation current Is, so on the middle layer the triple tip array can measure the fluctuations driven radial particle flux, and similar to the forked probe the Reynolds stress is measured by tip1, tip2 and tip3.

In conclusion, this paper addresses a recently arisen application of the Langmuir probe on fusion plasma diagnostics: direct measurement of zonal flows. Although the Langmuir probe is one of the oldest diagnostics, their theories and experimental techniques are still in progress. Using electrical probes to detect fluctuations does not involve the problems in the Langmuir probe theory, primarily lying in the interpretation of the electron-side probe voltage-current characteristics in strong magnetic fields, therefore with an elaborately designed probe array configuration, the Langmuir probe can present the unique diagnostic function in a series of dynamic processes related to zonal flows. Since the zonal flow dynamics is now widely believed crucial to plasma confinement physics, this new application of the Langmuir probe has already contributed and will continue contributing to this important field.

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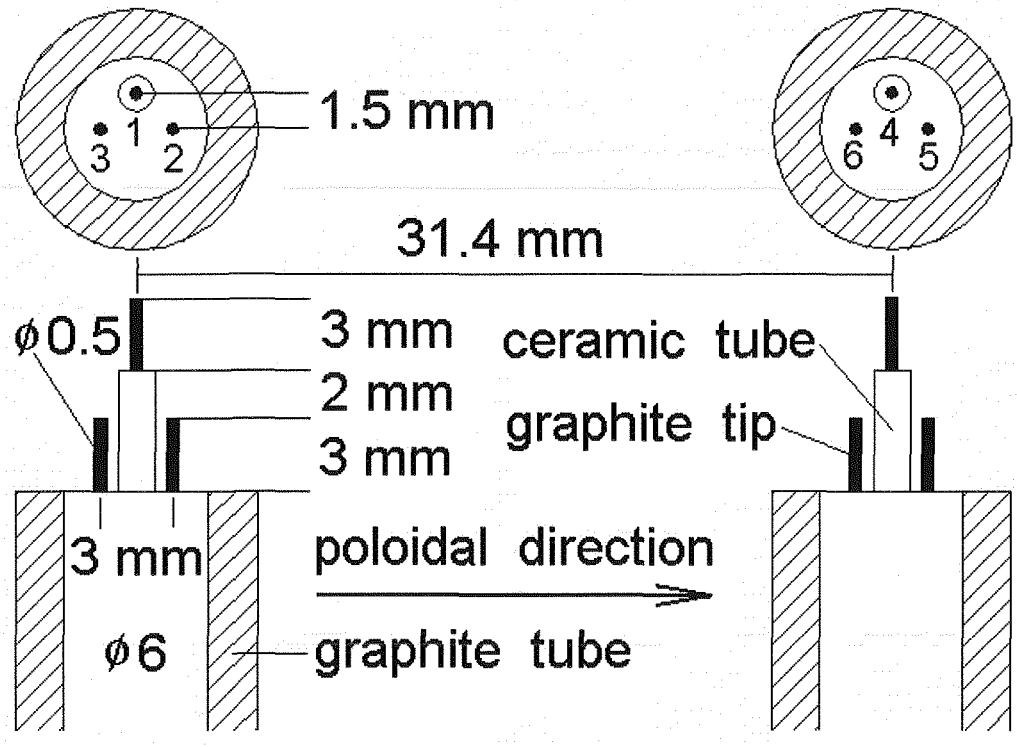


Figure 1 the configuration of forked probe array.

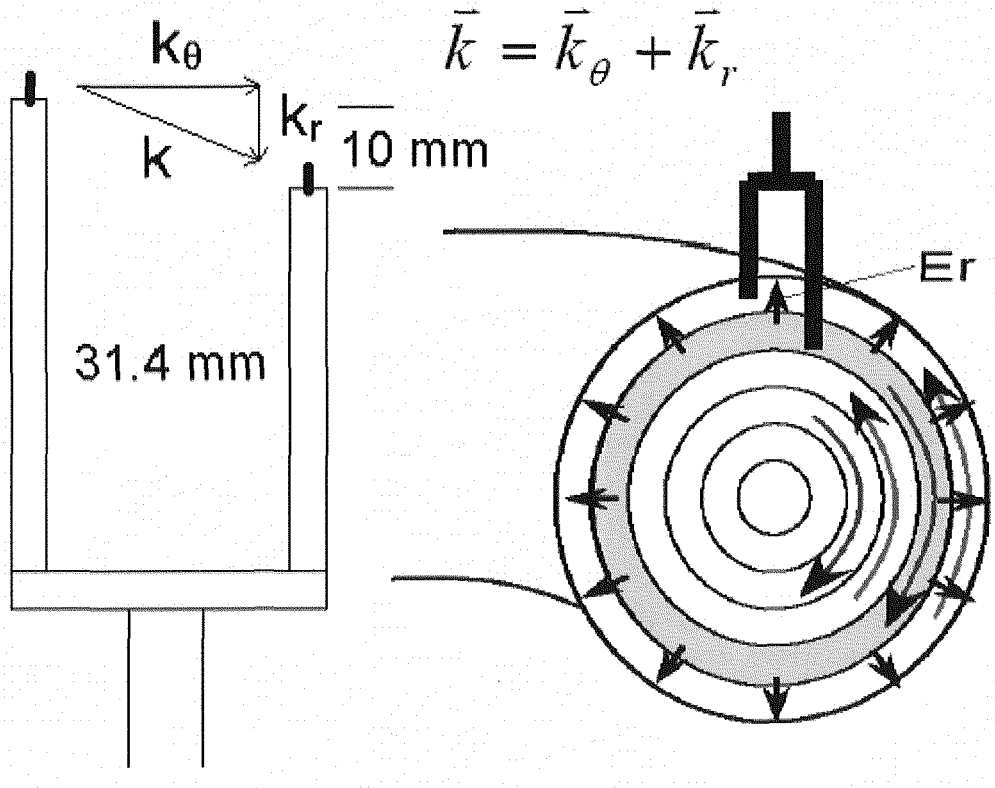


Figure 2 the scheme of k_r measuring principle using the forked probe array.

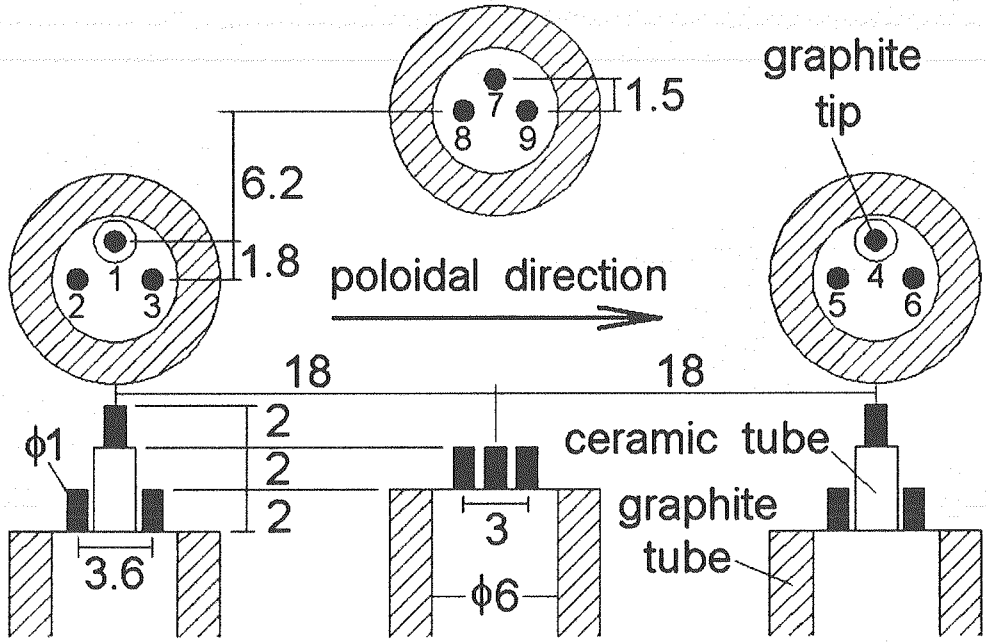


Figure 3 the configuration of farmer fork probe array.

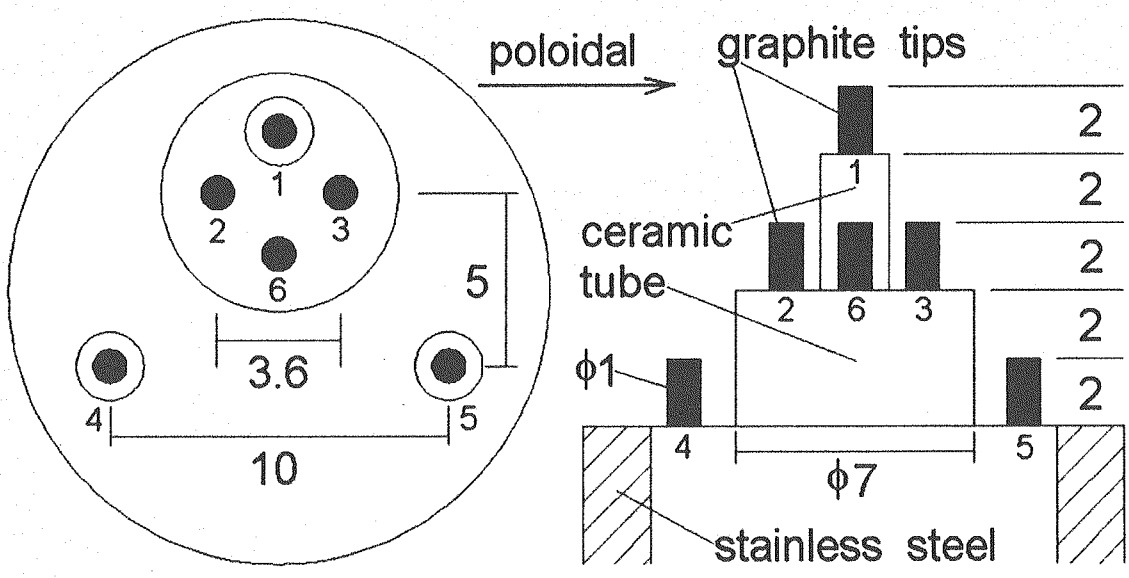


Figure 4 the configuration of pyramid probe array.

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